Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

In how many ways 5 identical balls can be distributed into 3 different boxes so that no box remains empty?

(A) 36

(B) 18

(C) 6

(D) 12

Q.2

Find the number of permutation of 4 letters taken from the word EXAMINATION.

(A) 1504

(B) 2520

(C) 2552

(D) 2454

Q.3

The sum of all numbers which can be formed with digits 1,2 and 3 is-

(A) 716

(B) 1432

(C) 2148

(D) None of these

Q.4

The number of ways in which 7 girls can be stand in a circle so that they do not have the same neighbour in any two arrangements?

(A) 720

(B) 380

(C) 360

(D) None of these

Q.5

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The number of ways in which 7 men and 7 women can sit on a circular table so that no two women sit together is

(A) $7!.7!$

(B) $7!.6!$

(C) $(6!)^2$

(D) $7!$

Q.6

There are four balls of different colours and four boxes of colours same as those of the balls. The number of ways in which the balls, one each box, could be placed such that a ball does not go to box of its own colour is-

(A) 8

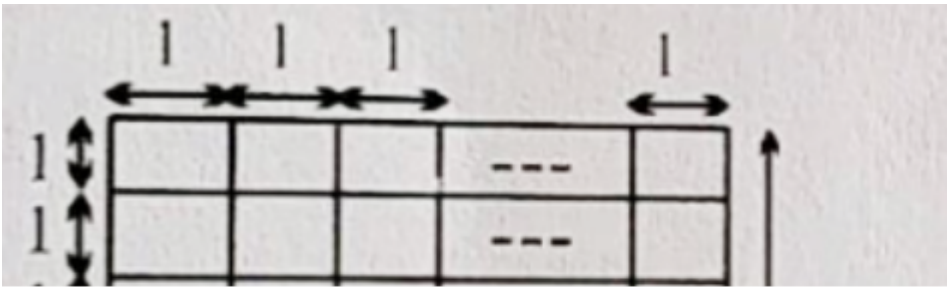
(B) 7

(C) 9

(D) None of these

Q.7

A rectangle with sides $2m-1, 2n-1$ is divided into squares of unit length by drawing parallel lines as shown in the diagram.
The number of rectangles with odd side length is



(A) $(m+n+1)^2$

(B) $mn(m+1)(n+1)$

(C) m^2n^2

(D) 4^{m+n-1}

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Q.8

Find the number of 6 digit numbers that can be made with the digits $1, 2, 3$ and 4 and having two pairs of digits.

(A) 1000

(B) 1080

(C) 980

(D) 1240

Q.9

Kanchan has 10 friends among whom two are married to each other. She wishes to invite five of them for a party. If the married couples refuse to attend separately, then the number of different ways in which she can invite five friends is -

(A) ${}^{8}\mathrm{C}_{5}$

(B) $2 \times {}^{8}\mathrm{C}_{3}$

(C) ${}^{10}\mathrm{C}_{5} - 2 \times {}^{8}\mathrm{C}_{4}$

(D) None of these

Q.10

If a seven-digit number made up of all distinct digits $8, 7, 6, 4, 3, x$ and y is divisible by 3 , then -

(A) maximum value of $x - y$ is 7

(B) maximum value of $x + y$ is 14

(C) maximum value of xy is 42

(D) maximum value of $x + y$ is 12

Q.11

Let n denote the number of ways of selecting 15 tickets from unlimited number of white, green, red and yellow tickets, such that each selection contains at least 3 white and 4 yellow tickets. Then n is

(A) divisible by 11

(B) divisible by 3

(C) divisible by 2

(D) divisible by 8

Q.12

Passage:
There are m seats in the first row of a theatre, of which n are to be occupied.

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Question:

The number of ways of arranging n persons if no two persons sit side by side is -

(A) $\frac{(\mathrm{m}-\mathrm{n}+1)!}{(\mathrm{m}-3\mathrm{n}+1)!}$

(B) $\frac{(\mathrm{m}-\mathrm{n}+1)!}{(\mathrm{m}-2\mathrm{n})!}$

(C) $\frac{(m-n+1)!}{(m-2n+1)!}$

(D) $\frac{(\mathrm{m}-\mathrm{n}+2)!}{(\mathrm{m}-2\mathrm{n}-1)!}$

Q.13

Passage:

There are m seats in the first row of a theatre, of which n are to be occupied.

Question:

If n is even, the number of ways of arranging n persons if each person has exactly one neighbour is

(A) $\left({}^{\mathrm{n}}\mathrm{P}_{\mathrm{n}/2}\right)\left({}^{\mathrm{m}-\mathrm{n}+1}\mathrm{P}_{\mathrm{n}/2}\right)$

(B) $\left({}^{n}P_{n}\right)\left({}^{m-n+1}P_{n/2}\right)$

(C) $\left({}^{\mathrm{n}}\mathrm{P}_{\mathrm{n}/2}\right)\left({}^{\mathrm{m}-\mathrm{n}+1}\mathrm{P}_{\mathrm{n}}\right)$

(D) None of these

Q.14

Passage:

There are m seats in the first row of a theatre, of which n are to be occupied.

Question:

The number of ways of arranging n persons, if out of any two seats located symmetrically in the middle of the row at least one is empty is -

(A) $\left({}^{m/2}C_{n}\right)\left(2^{n}\right)-1$

(B) ${}^{m/2}P_{n}$

(C) $\left({}^{m/2}\mathrm{P}_{\mathrm{n}}\right)\left(2^{\mathrm{n}}-1\right)$

(D) $\left({}^{m/2}\mathrm{P}_{\mathrm{n}}\right)\left(2^{\mathrm{n}}\right)$

Q.15

Passage:

Twelve 12 persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only.

Question:

Number of ways in which these 12 persons can be arranged is -

(A) ${}^{12}\mathrm{C}_{5}6!$

(B) $6!4!$

(C) ${}^{12}\mathrm{C}_{5}6!4!$

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(D) None of these

Q.16

Passage:
Twelve 12 persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only.

Question:
Number of ways of arrangements if two particular persons A and B do not want to be on the same table is -

(A) $^{10}C_4 6!4!$

(B) $2^{10}C_6 6!4!$

(C) $^{11}C_6 6!4!$

(D) None of these

Q.17

Passage:
Twelve 12 persons are to be arranged around two round tables such that one table can accommodate seven persons and another five persons only.

Question:
Numbers of ways of arrangement if two particular person A and B want to be together and consecutive is -

(A) $^{10}C_7 6!3!2! + {}^{10}C_5 4!5!2!$

(B) $^{10}C_5 6!3! + {}^{10}C_7 4!5!$

(C) $^{10}C_7 6!2! + {}^{10}C_5 5!2!$

(D) None of these

Q.18

Match the column -

Column I	**Column II**
(A) Four dice (six faced) are rolled. The number of possible outcomes in which at least one dice shows 2 is	(p) 210
(B) Let A be the set of 4-digit number $a_1a_2a_3a_4$ where $a_1 > a_2 > a_3 > a_4$. Then n (A) is equal to	(q) 480
(C) The total number of three-digit numbers, the sum of whose digits is even, is equal to	(r) 671
(D) The number of four-digit numbers that can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7 so that each number contains digit 1 is	(s) 450

(A) (A) - r, (B) - p, (C) - s, (D) - q

(B) (A) - s, (B) - q, (C) - r, (D) - p

(C) (A) - q, (B) - s, (C) - p, (D) - r

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(D) (A) - p, (B) - r, (C) - q, (D) - s

Q.19

If $6p^2$ numbers of 4 digits can be made with the digit 0, 1, 2, 3, 4, 5, which are divisible by 3, digit being unrepeated in the same number, then find the value of p + 4.

Q.20

Out of 5 apples, 10 mangoes and 15 oranges, the number of ways of distributing 15 fruits each to two persons, is

Answers & Solutions

Q.1 Answer:

6

Solution:

The required number of ways $= {}^{5-1}C_{3-1} = 4C_2 = \frac{4.3}{1.2} = 6$

Q.2 Answer:

2454

Solution:

Number of permutations are : (AA), (I I), (NN), E, X, M, T, O

$=$ **Coefficient of** x^4 **in** $4!\left(1+\frac{x}{1!}+\frac{x^2}{2!}\right)^3\left(1+\frac{x}{1!}\right)^5 =$ **Coefficient of** x^4 **in** $4!\left(1+x+\frac{x^2}{2!}\right)^3(1+x)^5$

$=$ **Coefficient of** x^4 **in** $4!\left\{(1+x)^3+\frac{x^6}{8}+\frac{3}{2}(1+x)^2x^2+\frac{3}{4}x^4(1+x)\right\}\cdot(1+x)^5$

$=$ **Coefficient of** x^4 **in** $4!\left\{(1+x)^8+\frac{x^6}{8}(1+x)^5+\frac{3}{2}x^2\left(1+x^7\right)+\frac{3}{4}x^4(1+x)^6\right\}$

$= 4!\left\{{}^8C_4+0+\frac{3}{2}{}^7C_2+\frac{3}{4}\right\} = 24\left\{\frac{8.7.6.5}{1.2.3.4}+\frac{3}{2}+\frac{7.6}{1.2}+\frac{34}{3}\right\}$

$= 8.7.6.5+6(3.7.6)+6.3 = 1680+756+18 = 2454$

Q.3 Answer:

None of these

Solution:

The sum of the digits $= 1+2+3 = 6$ **and** $n = 3$**, so the sum of all numbers formed** $= 6 \cdot 2!(111) = 12 \times 111 = 1332$

Q.4 Answer:

360

Solution:

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Seven girls can keep stand in a circle by $\frac{(7-1)!}{2!}$ number of ways, because there is no difference in anticlockwise and clockwise order of their standing in a circle.

$\therefore \quad \frac{(7-1)!}{2!} = 360$

Q.5 Answer:

$7!.6!$

Solution:

Here one women will sit between two men. Now fixing the place of one man the remaining 6 men on the circular table can sit in $6!$ ways. Since there are seven places between 7 men. Therefore seven women can sit on these places in $7!$ ways.

Thus 7 men and 7 women under the given condition can sit in $7!.6!$ ways.

Q.6 Answer:

9

Solution:

Number of derangements are $= 4!\left\{\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right\} = 12 - 4 + 1 = 9$

Since number of derangements in such a problems is given by $n!$

$\left\{1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \ldots (-1)^{\mathrm{n}} \frac{1}{\mathrm{n}!}\right\}$

Q.7 Answer:

m^2n^2

Solution:

If l is the length and b is the breadth of a rectangle, then $l \in \{1, 3, 5, \ldots, 2m-1\}$ and $b \in \{1, 3, 5, \ldots, 2n-1\}$

The number of rectangles of all possible sizes is $(1 + 3 + 5 + \ldots + (2m - 1))\left((1 + 3 + 5 + \ldots + (2n - 1)\right) = m^2n^2$

Q.8 Answer:

1080

Solution:

The number will have 2 pairs and 2 different digits

The number of selections $= {}^4\mathrm{C}_2 \times {}^2\mathrm{C}_2$**, and for each selection, number of arrangements** $= \frac{6!}{2!2!}$

Therefore, the required number of number $= {}^4C_2 \times {}^2C_2 \times \frac{6!}{2!2!} = 1080$

Q.9 Answer:

$2 \times {}^8\mathrm{C}_3$

Solution:

The number of ways of inviting, with the couple not included, is ${}^8\mathrm{C}_5$**.**

The number of ways of inviting with the couple included, is ${}^8\mathrm{C}_3$**.**

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Therefore, the required number of ways is

$^8C_5 + {}^8C_3 = {}^8C_3 + {}^8C_3 \quad (\because {}^8C_5 = {}^8C_3)$

Also, $^{10}C_5 - 2 \times {}^8C_4 = \frac{10!}{5!5!} - 2 \times \frac{8!}{4!4!}$

$= \frac{10\times9\times8\times7\times6}{120} - 2 \times \frac{8\times7\times6\times5}{24} = 9 \times 4 \times 7 - 140 = 112 = 2 \times \frac{8!}{3!5!}$

Q.10 Answer:

maximum value of $x - y$ is 7

Solution:

$8, 7, 6, 4, 3, x$ **and** y

Any number is divisible by 3 if sum of digits is divisible by 3 i.e, $x + y + 28$ **in divisible by 3**

A all digit are distinct $80x, 4y$ **can taken from** $0, 1, 2, 5, 9$

so passible pair (x, y) **can be**

$(0, 2), \quad (2, 0)$

$(0, 5), \quad (5, 0)$

$(2, 9), \quad (9, 2)$

$(5, 9), \quad (9, 5)$

maximum $x - y = 9 - 2 = 7$

maximum $x + y = 9 + 5 = 14$

maximum $xy = 9.5 = 45$

- No option match

Q.11 Answer:

divisible by 11

Solution:

The required number of ways

$=$ **coefficient of** t^{15} **in**

$\left(t^3 + t^4 \ldots\ldots\ldots\right)\left(t^0 + t + t^2 \ldots\ldots\ldots\right)\left(t^0 + t + t^2 \ldots\ldots\ldots\right)\left(t^4 + t^5 \ldots\ldots\infty\right)$

$=$ **coefficient of** t^8 **in** $\left(1 + t + t^2 \ldots\ldots\ldots\right)^4$

$=$ **coefficient of** t^8 **in** $(1 - t)^{-4} = {}^{(4+8-1)}C_8 = {}^{11}C_8 = {}^{11}C_3 = 11 \times 5 \times 3$

Q.12 Answer:

$\frac{(m-n+1)!}{(m-2n+1)!}$

Solution:

(i) Let x_0 **denote the number of empty seats to the left of the first person,** $x_i (1 \le i \le n-1)$ **the number of empty seats between the** i^{th} **and** $(i+1)^{\text{th}}$ **persons of the** n^{th} **person.**

Then $x_0, x_n \ge 0$ **and** $x_i \ge 1$ **for** $1 \le i \le n-1$

$\mathrm{x_0 + x_1 + \ldots\ldots + x_n = (m - n)}$ **....(1)**

Putting $x_i = y_i + 1$**, where** $y_i \ge 0$**, we have**

$\mathrm{x_0 + y_1 + \ldots\ldots\ldots + y_{n-1} + x_n + (1 + 1 + 1 + \ldots\ldots + (n-1) \text{ times }) = (m - n)}$

$\Rightarrow x_0 + y_1 + \ldots\ldots + y_{n-1} + x_n = m - n - (n-1)$

$\Rightarrow x_0 + y_1 + \ldots\ldots + y_{n-1} + x_n = m - 2n + 1$

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Now number of non-negative integral solutions is $\mathrm{n}+1+(\mathrm{m}-2\mathrm{n}+1)-1\mathrm{C}_{\mathrm{n}+1-1}=\mathrm{m}-\mathrm{n}+1\mathrm{C}_{\mathrm{n}}$.

Since we can permute n **persons in** $n!$ **ways, the required number of ways is**

$(\mathrm{m}-\mathrm{n}+1\mathrm{C}_{\mathrm{n}})\,(\mathrm{n}!)=\frac{(\mathrm{m}-\mathrm{n}+1)!}{\mathrm{n}!(\mathrm{m}-2\mathrm{n}+1)!}\mathrm{n}!=\frac{(\mathrm{m}-\mathrm{n}+1)!}{(\mathrm{m}-2\mathrm{n}+1)!}$

(ii) Let $\mathrm{n}=2\mathrm{k}$**, where** k **is some positive integer.**

Let x_0 **denote the number of empty seats to the left of the first pair,** $x_i(1\le i\le k-1)$ **the number of empty seats between** i^{th} **and** $(i+1)^{\mathrm{th}}$ **pair and** x_k **the number of empty seats to the right of the** k^{th} **pair.**

Note that $\mathrm{x}_0,\mathrm{x}_{\mathrm{k}}\ge 0,\mathrm{x}_{\mathrm{i}}\ge 1(1\le \mathrm{i}\le \mathrm{k}-1)$ **and** $\mathrm{x}_0+\mathrm{x}_1+\ldots\ldots\ldots\ldots+\mathrm{x}_{\mathrm{k}}=(\mathrm{m}-2\mathrm{k})$ **(2)**

The number of integral solutions of eq. (2) is $\mathrm{m}-2\mathrm{k}+1\mathrm{C}_{\mathrm{k}}$.

Since we can permute n **persons in** $n!$ **ways, the required number of ways is**

$$(\mathrm{m}-2\mathrm{k}+1\mathrm{C}_{\mathrm{k}})\,(2\mathrm{k})!=\frac{(\mathrm{m}-2\mathrm{k}+1)!}{\mathrm{k}!(\mathrm{m}-3\mathrm{k}+1)!}(2\mathrm{k})!=\frac{(2\mathrm{k})!(\mathrm{m}-2\mathrm{k}+1)!}{(\mathrm{k}!)(\mathrm{m}-3\mathrm{k}+1)!}$$
$$=\left({}^{2\mathrm{k}}\mathrm{P}_{\mathrm{k}}\right)\left({}^{\mathrm{m}-2\mathrm{k}+1}\mathrm{P}_{\mathrm{k}}\right)=\left({}^{\mathrm{n}}\mathrm{P}_{\mathrm{n}/2}\right)\left({}^{\mathrm{m}-\mathrm{n}+1}\mathrm{P}_{\mathrm{n}/2}\right)$$

(iii) m **is even. Let** $\mathrm{m}=2\mathrm{k}$**, where** k **is some positive integer.**

We can choose n **seats out of the** k **seats to the left of the middle seat in** k_{C} **ways.**

Each chosen seat can be either empty or occupied.

Thus, the number of ways of choosing seats for n persons is equal to $\left({}^{k}C_{n}\right)\left(2^{n}\right)$.

We can arrange n **persons at these seats in** ${}^{n}P_{n}$ **ways.**

Hence, the required number of arrangements is given by $(\mathrm{n}!)\left({}^{\mathrm{k}}\mathrm{C}_{\mathrm{n}}\right)\left(2^{\mathrm{n}}\right)=\left(\mathrm{kP}_{\mathrm{n}}\right)\left(2^{\mathrm{n}}\right)=\left({}^{\mathrm{m}/2}\mathrm{P}_{\mathrm{n}}\right)\left(2^{\mathrm{n}}\right)$

Q.13 Answer:

$\left({}^{\mathrm{n}}\mathrm{P}_{\mathrm{n}/2}\right)\left({}^{\mathrm{m}-\mathrm{n}+1}\mathrm{P}_{\mathrm{n}/2}\right)$

Solution:

(i) Let x_0 **denote the number of empty seats to the left of the first person,** $x_i(1\le i\le n-1)$ **the number of empty seats between the** i^{th} **and** $(i+1)^{\mathrm{th}}$ **persons of the** n^{th} **person.**

Then $x_0,x_n\ge 0$ **and** $x_i\ge 1$ **for** $1\le i\le n-1$

$\mathrm{x}_0+\mathrm{x}_1+\ldots\ldots+\mathrm{x}_{\mathrm{n}}=(\mathrm{m}-\mathrm{n})$ **....(1)**

Putting $x_i=y_i+1$**, where** $y_i\ge 0$**, we have**

$\mathrm{x}_0+\mathrm{y}_1+\ldots\ldots\ldots+\mathrm{y}_{\mathrm{n}-1}+\mathrm{x}_{\mathrm{n}}+(1+1+1+\ldots\ldots+(\mathrm{n}-1)\text{ times })=(\mathrm{m}-\mathrm{n})$

$\Rightarrow x_0+y_1+\ldots\ldots+y_{n-1}+x_n=m-n-(n-1)$

$\Rightarrow x_0+y_1+\ldots\ldots+y_{n-1}+x_n=m-2n+1$

Now number of non-negative integral solutions is $\mathrm{n}+1+(\mathrm{m}-2\mathrm{n}+1)-1\mathrm{C}_{\mathrm{n}+1-1}=\mathrm{m}-\mathrm{n}+1\mathrm{C}_{\mathrm{n}}$.

Since we can permute n **persons in** $n!$ **ways, the required number of ways is**

$(\mathrm{m}-\mathrm{n}+1\mathrm{C}_{\mathrm{n}})\,(\mathrm{n}!)=\frac{(\mathrm{m}-\mathrm{n}+1)!}{\mathrm{n}!(\mathrm{m}-2\mathrm{n}+1)!}\mathrm{n}!=\frac{(\mathrm{m}-\mathrm{n}+1)!}{(\mathrm{m}-2\mathrm{n}+1)!}$

(ii) Let $\mathrm{n}=2\mathrm{k}$**, where** k **is some positive integer.**

Let x_0 **denote the number of empty seats to the left of the first pair,** $x_i(1\le i\le k-1)$ **the number of empty seats between** i^{th} **and** $(i+1)^{\mathrm{th}}$ **pair and** x_k **the number of empty seats to the right of the** k^{th} **pair.**

Note that $\mathrm{x}_0,\mathrm{x}_{\mathrm{k}}\ge 0,\mathrm{x}_{\mathrm{i}}\ge 1(1\le \mathrm{i}\le \mathrm{k}-1)$ **and** $\mathrm{x}_0+\mathrm{x}_1+\ldots\ldots\ldots\ldots+\mathrm{x}_{\mathrm{k}}=(\mathrm{m}-2\mathrm{k})$ **(2)**

The number of integral solutions of eq. (2) is $\mathrm{m}-2\mathrm{k}+1\mathrm{C}_{\mathrm{k}}$.

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Since we can permute n persons in $n!$ ways, the required number of ways is

$$(m-2k+1C_k)\,(2k)! = \frac{(m-2k+1)!}{k!(m-3k+1)!}(2k)! = \frac{(2k)!(m-2k+1)!}{(k!)(m-3k+1)!}$$
$$= \left({}^{2k}P_k\right)\left({}^{m-2k+1}P_k\right) = \left({}^{n}P_{n/2}\right)\left({}^{m-n+1}P_{n/2}\right)$$

(iii) m is even.

Let $m = 2k$, where k is some positive integer.

We can choose n seats out of the k seats to the left of the middle seat in kC ways.

Each chosen seat can be either empty or occupied.

Thus, the number of ways of choosing seats for n persons is equal to $\left({}^{k}C_n\right)(2^n)$.

We can arrange n persons at these seats in ${}^{n}P_n$ ways.

Hence, the required number of arrangements is given by $(n!)\left({}^{k}C_n\right)(2^n) = (kP_n)\,(2^n) = \left({}^{m/2}P_n\right)(2^n)$

Q.14 Answer:

$\left({}^{m/2}P_n\right)(2^n)$

Solution:

(i) Let x_0 denote the number of empty seats to the left of the first person, $x_i (1 \le i \le n-1)$ the number of empty seats between the ith and $(i+1)^{th}$ persons of the nth person.

Then $x_0, x_n \ge 0$ and $x_i \ge 1$ for $1 \le i \le n-1$

$$x_0 + x_1 + \ldots\ldots + x_n = (m-n) \text{ ....(1)}$$

Putting $x_i = y_i + 1$, where $y_i \ge 0$, we have

$$x_0 + y_1 + \ldots\ldots + y_{n-1} + x_n + (1+1+1+\ldots\ldots + (n-1) \text{ times }) = (m-n)$$
$$\Rightarrow x_0 + y_1 + \ldots\ldots + y_{n-1} + x_n = m - n - (n-1)$$
$$\Rightarrow x_0 + y_1 + \ldots\ldots + y_{n-1} + x_n = m - 2n + 1$$

Now number of non-negative integral solutions is $n+1+(m-2n+1)-1C_{n+1-1} = m-n+1C_n$.

Since we can permute n persons in $n!$ ways, the required number of ways is

$$(m-n+1C_n)\,(n!) = \frac{(m-n+1)!}{n!(m-2n+1)!}n! = \frac{(m-n+1)!}{(m-2n+1)!}$$

(ii) Let $n = 2k$, where k is some positive integer.

Let x_0 denote the number of empty seats to the left of the first pair, $x_i (1 \le i \le k-1)$ the number of empty seats between i^{th} and $(i+1)^{th}$ pair and x_k the number of empty seats to the right of the kth pair.

Note that $x_0, x_k \ge 0, x_i \ge 1 (1 \le i \le k-1)$ and $x_0 + x_1 + \ldots\ldots\ldots\ldots + x_k = (m-2k)$ (2)

The number of integral solutions of eq. (2) is $m-2k+1C_k$.

Since we can permute n persons in $n!$ ways, the required number of ways is

$$(m-2k+1C_k)\,(2k)! = \frac{(m-2k+1)!}{k!(m-3k+1)!}(2k)! = \frac{(2k)!(m-2k+1)!}{(k!)(m-3k+1)!}$$
$$= \left({}^{2k}P_k\right)\left({}^{m-2k+1}P_k\right) = \left({}^{n}P_{n/2}\right)\left({}^{m-n+1}P_{n/2}\right)$$

(iii) m is even.

Let $m = 2k$, where k is some positive integer.

We can choose n seats out of the k seats to the left of the middle seat in kC ways.

ach chosen seat can be either empty or occupied.

Thus, the number of ways of choosing seats for n persons is equal to $\left({}^{k}C_n\right)(2^n)$.

We can arrange n persons at these seats in ${}^{n}P_n$ ways.

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Hence, the required number of arrangements is given by $(n!)\left({}^{k}C_{n}\right)\left(2^{n}\right)=\left(kP_{n}\right)\left(2^{n}\right)=\left({}^{m/2}P_{n}\right)\left(2^{n}\right)$

Q.15 Answer:

${}^{12}C_{5}6!4!$

Solution:

(i) Seven persons can be selected for first table in ${}^{12}C_{7}$ **ways.**

Now these seven persons can be arranged in $6!$ **ways.**

The remaining five persons can be arranged on the second table in $4!$ **ways.**

Hence, total number of ways is ${}^{12}C_{5}6!4!$.

(ii) Here, A **can sit on first table** B **on the second or** A **on second table and** B **on the second table.**

If A **is on the first table, then remaining six for first table can be selected in** ${}^{10}C_{6}$ **ways.**

Now these seven persons can be arranged in $6!$ **ways.**

Remaining five can be arranged on the other table in $4!$ **ways.**

Hence, total number of ways is $2{}^{10}C_{6}6!4!$.

(iii) If A, B **are on the first table, then remaining five can be selected in** ${}^{10}C_{5}$ **ways.**

Now seven persons including A **and** B **can be arranged on the first table in which** A **and** B **are together in** $2!5!$ **ways.**

Remaining five can be arranged on the second table in $4!$ **ways.**

Total number of ways is ${}^{10}C_{5}4!5!2!$

If A, B **are on the second table, then remaining three can be selected in** ${}^{10}C_{3}$ **ways.**

Now five persons including A **and** B **can be arranged on the second table in which** A **and** B **are together** $2!3!$ **ways.**

Remaining seven can be arranged on the first table in $6!$ **ways.**

Hence, number of ways for first table is ${}^{10}C_{7}6!3!2!$.

Q.16 Answer:

$2{}^{10}C_{6}6!4!$

Solution:

(i) Seven persons can be selected for first table in ${}^{12}C_{7}$ **ways.**

Now these seven persons can be arranged in $6!$ **ways.**

The remaining five persons can be arranged on the second table in $4!$ **ways.**

Hence, total number of ways is ${}^{12}C_{5}6!4!$.

(ii) Here, A **can sit on first table** B **on the second or** A **on second table and** B **on the second table.**

If A **is on the first table, then remaining six for first table can be selected in** ${}^{10}C_{6}$ **ways.**

Now these seven persons can be arranged in $6!$ **ways.**

Remaining five can be arranged on the other table in $4!$ **ways.**

Hence, total number of ways is $2{}^{10}C_{6}6!4!$.

(iii) If A, B **are on the first table, then remaining five can be selected in** ${}^{10}C_{5}$ **ways.**

Now seven persons including A **and** B **can be arranged on the first table in which** A **and** B **are together in** $2!5!$ **ways.**

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Remaining five can be arranged on the second table in $4!$ ways.

Total number of ways is ${}^{10}\mathrm{C}_5 4!5!2!$

If A, B are on the second table, then remaining three can be selected in ${}^{10}C_3$ ways.

Now five persons including A and B can be arranged on the second table in which A and B are together $2!3!$ ways.

Remaining seven can be arranged on the first table in $6!$ ways.

Hence, number of ways for first table is ${}^{10}\mathrm{C}_7 6!3!2!$.

Q.17 Answer:

${}^{10}\mathrm{C}_7 6!3!2! + {}^{10}\mathrm{C}_5 4!5!2!$

Solution:

(i) Seven persons can be selected for first table in ${}^{12}C_7$ ways.

Now these seven persons can be arranged in $6!$ ways.

The remaining five persons can be arranged on the second table in $4!$ ways.

Hence, total number of ways is ${}^{12}\mathrm{C}_5 6!4!$.

(ii) Here, A can sit on first table B on the second or A on second table and B on the second table.

If A is on the first table, then remaining six for first table can be selected in ${}^{10}\mathrm{C}_6$ ways.

Now these seven persons can be arranged in $6!$ ways.

Remaining five can be arranged on the other table in $4!$ ways.

Hence, total number of ways is $2{}^{10}\mathrm{C}_6 6!4!$.

(iii) If A, B are on the first table, then remaining five can be selected in ${}^{10}\mathrm{C}_5$ ways.

Now seven persons including A and B can be arranged on the first table in which A and B are together in $2!5!$ ways.

Remaining five can be arranged on the second table in $4!$ ways.

Total number of ways is ${}^{10}\mathrm{C}_5 4!5!2!$

If A, B are on the second table, then remaining three can be selected in ${}^{10}C_3$ ways.

Now five persons including A and B can be arranged on the second table in which A and B are together $2!3!$ ways.

Remaining seven can be arranged on the first table in $6!$ ways.

Hence, number of ways for first table is ${}^{10}\mathrm{C}_7 6!3!2!$.

Q.18 Answer:

(A) - r, (B) - p, (C) - s, (D) - q

Solution:

$(\mathrm{A}) - \mathbf{r}, (\mathrm{B}) - \mathbf{p}, (\mathbf{C}) - \mathbf{s}, (\mathrm{D}) - \mathbf{q}$

(A) The number of possible outcomes with 2 on at least one dice $=$ The total number of outcomes with 2 on at least one dice

$=$ (The total number of outcomes) $-$ (The number of outcomes in which 2 does not appear on any dice)

$= 6^4 - 5^4 = 1296 - 625 = 671$

(B) Any selection of four digits from the 10 digits $0, 1, 2, 3, \ldots\ldots, 9$ gives one number.

So, the required number of numbers is ${}^{10}\mathrm{C}_4$.

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(C) Let the number be $\mathrm{n} = \mathrm{pqr}$.

Since $\mathrm{p} + \mathrm{q} + \mathrm{r}$ is even, p can be filled in 9 ways and q can be filled in 10 ways.

r can be filled in number of ways depending upon what is the sum of p and q.

If $p + q$ is odd, the r can be filled with any one of five odd digits.

If $p + q$ is even, then r can be filled with any one of five even digits.

In any case, r can be filled in five ways.

Hence, total number of numbers is $9 \times 10 \times 5 = 450$

(D) After fixing 1 at one position out of 4 places 3 places can be filled by ${}^{7}\mathrm{P}_{3}$ ways.

But for some number whose fourth digit is zero, such type of ways is $6\mathrm{P}_{2}$.

Therefore, total number ways is ${}^{7}\mathrm{P}_{3} - {}^{6}\mathrm{P}_{2} = 480.$

Q.19 Answer:

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Solution:

A number of divisible by 3 If its sum of digits are divisible by 3.

So the number can be formed by using the digits.

0, 1, 2, 3 or 0, 2, 3, 4 or 0, 1, 3, 5.

or 1, 2, 4, 5 or 0, 3, 4, 5

Number of 4 digits number divisible by 3

= 4[(4! – 3!)] + 4! = 96 = 6p^2

$\therefore$ p = 4 and so p + 4 = 8

Q.20 Answer:

—

Solution:

The desired number is the coefficient of x^{15} in the product

$$\left(1+x+x^2+\ldots+x^5\right)\left(1+x+x^2+\ldots+x^{10}\right)\left(1+x+x^2+\ldots+x^{15}\right)$$
$$=\left(1-x^6\right)\left(1-x^{11}\right)\left(1-x^{16}\right)(1-x)^{-3}$$
$$=\left(1-x^6-x^{11}+\ldots\right)\left(1+\left(\begin{array}{c}3\\1\end{array}\right)x+\left(\begin{array}{c}4\\2\end{array}\right)x^2+\ldots\right)$$

which is $\left(\begin{array}{c}17\\15\end{array}\right)-\left(\begin{array}{c}11\\9\end{array}\right)-\left(\begin{array}{c}6\\4\end{array}\right)=\left(\begin{array}{c}17\\2\end{array}\right)-\left(\begin{array}{c}11\\2\end{array}\right)-\left(\begin{array}{c}6\\2\end{array}\right)=136-55-15=66.$

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